

Response to the Associate Editor for MS-OPM-2025-05030

We want to start by thanking you and your review team for the incredibly helpful feedback. We have been highly receptive to the comments, and worked diligently to ensure that the revised manuscript addresses every point raised by you and the three referees. Before responding point by point, we would like to highlight the most significant changes in the revised manuscript.

- *Added literature and positioning.* We have added a new paragraph in Section 1.2 discussing ?, ?, ?, the ? survey, and ?, together with a comparison table (Table 1). The paragraph articulates three points of distinction: (i) a different data structure (our auxiliary data do not observe the true outcome, while the cited papers observe outcomes in both sources); (ii) different identification assumptions (missing-at-random by design versus confounding correction); and (iii) a different primary contribution (optimal design of data collection versus estimation with given datasets).
- *Added empirical content.* We added Section 5.2, an ATE case study on the Hillstrom email marketing RCT (64,000 customers, three-arm design). INSPIRE-ATE achieves an average of +13.3% MSE reduction across all 8 experimental conditions, versus +4.2% for the Stratified/Neyman baselines. We also added Section 5.1.3, a robustness study across five predictor types (GPT-4o-mini, DeepSeek-V3.2, XGBoost, Population Mean, Random), finding 37–47% MSE reduction across all predictors and both datasets. The Census ML experiment has been moved to Appendix A, and guidance on the $\mathbf{X}/\mathbf{W}$ covariate partition has been added as Appendix C.
- *Improved writing.* Sections 2–3 have been reorganized; the narrative on bias correction and vanishing bias is now clarified throughout. A discussion of early-stage practical guidance has been added to Section 3.3. The manuscript has been proofread throughout, with all identified typos corrected.

AE Major Issue 1: Positioning and Distinction Relative to Existing Literature.

Reviewer 1 highlights substantial literature that studies bias correction and efficiency gains when combining high-quality experimental data with lower-quality auxiliary data. Although these papers typically do not focus on active learning, the conceptual overlap is nontrivial, especially in the paper’s non-adaptive setting where sampling and measurement rules are fixed.

I agree with Reviewer 1 on his/her points. It is very important to position the paper accurately, and revising with respect to their comments will strengthen the paper.

Our Response:

We agree that accurate positioning is essential, and we appreciate you and Reviewer 1 for drawing our attention to this literature. In the revised manuscript, we have added a thorough discussion of the works on combining experimental and observational data, including ?, ?, ?, ?, and ?. We have added a dedicated paragraph in the literature review (Section 1.2) to discuss the differences. We summarize the key distinctions below.

1. *Different data structure.* The cited papers study settings where a researcher has two separate data sources: a small experimental sample free of confounding and a large observational dataset with unmeasured confounders. The goal is to estimate the average treatment effect, and the key challenge is that the observational data is cannot be directly combined due to unmeasured confounding. On the other side, our setting is full experimental with no unmeasured confounders. The key challenge in our setting

is that some of the outcomes for units are missing, and we cannot directly use the prediction because of unknown bias in the prediction. We summarize the distinction in Table 1.

	Exp.+obs. literature		Our setting	
	Core data	Auxiliary data	Core data	Auxiliary data
Outcome observed?	Yes	Yes	Yes	No
Outcome unconfounded?	Yes	No	Yes	Yes
Same population?	No	No	Yes	Yes
Main challenge	Leverage auxiliary data that still observe the true outcome but may be confounded.		Leverage auxiliary data that are unconfounded by design but do not observe the true outcome.	

Table 1: Summary of the data-structure difference between the experimental-plus-observational data combination literature and our setting.

2. Different identification assumptions and estimation methods. These structural differences lead to distinct identification strategies. In the cited papers, the authors either use auxiliary observational data to estimate an outcome model and then apply it to the experimental data for precision improvement (e.g., ?), or use the experimental data to estimate confounding bias and then apply a control-function approach to debias the observational estimates (e.g., ?). In both cases, the core obstacle is confounding bias, and the estimation methods are designed to correct for it. Our method instead relies on the missing-at-random (MAR) assumption, which holds by design in our data collection framework. The MAR structure ensures that the outcome distribution is the same across the labeled and unlabeled subsets. Under this setup, we derive an orthogonal estimator that achieves the semiparametric efficiency bound. The bias correction in our estimator is not a control-function adjustment for confounding, but rather a correction for prediction error within a single population.

3. Optimal design of data collection and adaptive extension. The cited papers largely take the available datasets as given. By contrast, one of our main contributions is to optimize the data collection design itself. We derive the optimal data collection scheme in the non-adaptive setting and we further extend the framework to an adaptive setting where both the predictor and the design are updated sequentially as new labels arrive. To the best of our knowledge, this joint design problem and its adaptive extension have not been studied in the experimental/observational data combination literature.

AE Major Issue 2: Need for stronger empirical examples.

Reviewers 2 and 3 note that the paper would be strengthened by a more compelling empirical illustration, particularly one that showcases the framework’s ability to estimate average treatment effects, which is one of the paper’s most practically important extensions.

I would like to add that it would be particularly nice to see an empirical illustration for average treatment effect (ATE) estimation that showcases a concrete operations management application, especially given that the paper is submitted to the OM department. This is not a requirement for publication, but such an example would further increase the paper’s impact and accessibility for the OM audience. If developing an ATE illustration turns out to be substantially more involved than anticipated, an additional empirical example (ATE or not) with a clear OM application would also be valuable.

Our Response:

We agree that a concrete OM-relevant ATE illustration would substantially strengthen the paper, and we have added one as Section 5.2 in the revised manuscript.

The new case study uses the Hillstrom email marketing RCT, a publicly available dataset of 64,000 customers randomly assigned to one of three conditions: a men’s email campaign, a women’s email campaign, or no email. The outcomes are binary visit (did the customer visit the website within two weeks?) and continuous spend (dollars spent in the same period). The predictor is a gradient boosted outcome model trained on eight customer covariates (recency, frequency, monetary value, zip code, channel history, and three product-category indicators). This setting is directly relevant to operations management: firms routinely face the question of how to allocate a targeting or measurement budget across customer segments, and CATE estimation under a labeling budget is a natural formulation of that problem.

We report results for all eight cells of the design (two outcomes $\times$ two treatment contrasts $\times$ two budget levels) in Table 4. INSPIRE-ATE achieves a higher MSE reduction than all baselines in every cell. Averaging across conditions, INSPIRE-ATE reduces MSE by 13.3% relative to a naive design, while Stratified sampling and Neyman allocation achieve only 4.2%. The gain is especially pronounced for the spend outcome, where raw-covariate strata fail to concentrate measurement in informative subgroups, and INSPIRE’s prediction-guided design yields reductions of 9.0% to 19.9%, compared with often negative or near-zero gains for the baselines.

We have also moved the Census ML experiment (average income estimation with a decision-tree predictor) to Appendix A. The main empirical section now features the two experiments we regard as most compelling: the public-opinion survey case in Section 5.1 (with GPT-4o-mini, robustness checks, and an adaptive illustration) and the Hillstrom ATE case in Section 5.2.

Response to Reviewer 1 for MS-OPM-2025-05030

We thank Reviewer 1 for the careful reading and for bringing the related literature to our attention. Below we respond point by point.

R1.1: Paper summary.

This paper proposes a method for combining freely available (but potentially biased or erroneous) predictions with higher cost (but potentially higher quality) data. This method would allow practitioners to combine easy to obtain data, such as LLM predictions, with more costly data such as those obtained through A/B tests and further measurement.

Our Response:

We appreciate the reviewer’s accurate summary of our work.

R1.2: Related literature on combining experimental and observational data.

While the topic of combining high cost sampled data with freely available data no doubt unlocks value for modern use cases, there has been a large stream of work in this area working on very similar settings, that the existing work does not reference or build on.

One stream of work focuses on how to combine (plentiful) offline data with RCT data on smaller sample sizes. Some examples include Athey, Chetty, Imbens ’20, Ghassami et al ’22, and Gagnon-Bartsch ’23. (For more references, see Rosenman ’25 for a survey.) A key distinction between this stream of work and the current paper is that this stream of work largely does not consider active learning, which is the focus of the current paper. However, the key ideas overlap. In the current paper, the key idea is: “Given the two groups, we will utilize the data from the first group [with the true label and the prediction] to estimate the bias between the predictions and the true labels. Then, we would apply the estimated bias to the second group [with only predictions]... to infer the values of the true labels in that group.” The key idea from Athey et al ’20 is similar: “We interpret differences in the estimated causal effects on the secondary outcome between the two samples as evidence of unobserved confounders in the observational sample, and develop control function methods for using those differences to adjust the estimates of the treatment effects on the primary outcome.”

This above stream of work is very closely related to the non-adaptive setting that the authors first consider in Section 3 where they consider settings where both the measurement probability and predictors are static. While there is a distinction between the existing work that considers offline data and costly RCTs with the current paper that considers offline data and costly sampling, one would expect that these are closely related topics and indeed the authors extend their framework to costly RCTs in Section 4. While the above stream of work does not consider how to do active learning, which the current paper does, there may be similarities in the estimation that the authors need to consider.

Our Response (addressing both R1.2 and R1.3):

This is a very helpful comment. The high-level motivation is indeed shared: all of these papers, and ours, seek to leverage auxiliary lower-quality data to improve the efficiency of statistical inference. We have now added a thorough discussion of ?, ?, ?, and the ? survey to the revised manuscript, and we carefully articulate both the conceptual connections and the key differences.

The key difference lies in the data structure and the source of bias in the lower-quality data, which in turn lead to different identification assumptions and estimation methods. In short, our paper addresses bias from prediction error of an external model, while the cited papers address confounding bias in observational data. These are different problems, and the methods developed for one do not directly apply to the other. We elaborate below.

1. *The data structure and source of bias are different.* The cited papers study settings where a researcher has two separate data sources: a small experimental sample free of confounding and a large observational dataset with unmeasured confounders. The goal is to estimate the average treatment effect, and the key challenge is that the observational data cannot be naively combined with experimental data due to unmeasured confounding.

Our setting is different. Although we also have two groups of data, the outcomes of the lower-quality group are not observed at all; instead, only predictions from an external model are available. The two groups are not separate data sources but rather two subsets of a single experimental population, and the bias arises from prediction error rather than confounding. This stands in sharp contrast to the cited papers, where outcomes are observed in both data sources but the observational outcomes are confounded.

In short, our paper and the cited papers deal with different bias from the auxiliary dataset. Our paper deals with bias arise from imperfect predictions while the cited paper primarily handles bias due to unmeasured confoundings. We summarize the differences in Table 2.

	Exp.+obs. literature		Our setting	
	Core data	Auxiliary data	Core data	Auxiliary data
Outcome observed?	Yes	Yes	Yes	No
Outcome unconfounded?	Yes	No	Yes	Yes
Same population?	No	No	Yes	Yes
Main challenge	Leverage auxiliary data that still observe the true outcome but may be confounded.		Leverage auxiliary data that are unconfounded by design but do not observe the true outcome.	

Table 2: Summary of the data-structure difference between the experimental-plus-observational data combination literature and our setting.

2. *Different identification assumptions and estimation methods.* These structural differences lead to distinct identification strategies. In the cited papers, the authors either use auxiliary observational data to estimate an outcome model and then apply it to the experimental data for precision improvement (e.g., ?), or use the experimental data to estimate confounding bias and then apply a control-function approach to debias the observational estimates (e.g., ?). In both cases, the core obstacle is confounding bias, and the estimation methods are designed to correct for it.

Our method instead relies on the missing-at-random (MAR) assumption, which holds by design in our data collection framework. The MAR structure ensures that the outcome distribution is the same across the labeled and unlabeled subsets. Under this setup, we derive an orthogonal estimator that achieves the semiparametric efficiency bound. The bias correction in our estimator is not a control-function adjustment for confounding, but rather a correction for prediction error within a single population.

3. *Optimal design of data collection and adaptive extension.* The cited papers largely take the available datasets as given. By contrast, one of our main contributions is to optimize the data collection design

itself. We derive the optimal data collection scheme in the non-adaptive setting and we further extend the framework to an adaptive setting where both the predictor and the design are updated sequentially as new labels arrive. To the best of our knowledge, this joint design problem and its adaptive extension have not been studied in the experimental/observational data combination literature.

R1.3: Lykouris and Weng (2025) — adaptive setting with active learning.

Another very related paper to consider is Lykouris and Wang ('25), which incorporates both the adaptive setting of the current paper and the active learning component. This paper considers when to use cheap AI predictions versus when to require costly human labeling, where the human labels can also be used to train the AI predictions in future time steps. The constraints in Lykouris and Wang are different from the current paper, where the former incorporates cost through constrained supply and a queueing mechanism, whereas the latter incorporates a fixed budget constraint, but the high level idea is similar.

Our Response:

We appreciate this reference and have added it to the revised manuscript. Both papers share the high-level theme of deciding when to use cheap AI predictions versus costly human labels, but the settings, objectives, and methods are quite different.

? study an online queueing and classification problem. In their setting, the arrival process of items is exogenous, and a dispatcher decides per item whether to route it to a costly human expert or accept the AI prediction. The resource constraint is the human capacity and the key tradeoff is between classification accuracy and waiting-time costs as items queue up.

In contrast, we study a statistical estimation problem. The goal is to estimate a population-level quantity, either a mean or an average treatment effect, with maximum precision under a labeling budget. The data collection process is endogenous: the researcher chooses the sampling density and measurement probability across subpopulations. The resource constraint is a fixed labeling budget, and the key tradeoff is between budget expenditure and estimator variance. Our methods are rooted in semiparametric efficiency theory, whereas ? use online learning techniques for sequential classification decisions.

R1.5: Scope of contribution may be narrower than claimed.

It is very likely that the authors have made contributions in their own work beyond the contributions in the existing work. However, these distinctions must be carefully considered in future drafts of the work and it is possible that the scope of the contribution is more narrow, given the existing work.

Our Response:

Thank you for raising this point. After carefully reviewing all of the cited literature, we believe our contributions remain substantial but agree they must be stated precisely. We see three core contributions that differentiate our work from the papers the reviewer highlighted.

First, we formulate and solve the problem of jointly optimizing the sampling density $p(x)$ and measurement probability $\pi(x)$ to minimize estimator variance subject to a labeling budget. All cited works in the experimental/observational data combination literature take their datasets as given. The closest work on the design side is ?, who study active data collection in the prediction-powered inference framework. Our work differs from theirs in two respects. (i) ? optimize the measurement rule for a specific IPW-type estimator. Our approach instead minimizes the semiparametric efficiency lower bound, so the resulting design is optimal across the entire class of regular estimators. (ii) ? only optimize the measurement probability

$\pi(x)$, whereas we jointly optimize both the sampling density $p(x)$ and the measurement probability $\pi(x)$, which yields strictly larger efficiency gains as demonstrated in our numerical experiments.

Second, on the methodological side, we make two contributions that go beyond the cited papers. (i) We derive a new semiparametric efficiency bound under a statistical model where labels are MAR with designer-chosen missingness and an external predictor is available for all units. The bound in Theorem 1 is a function of the design variables $p(x)$ and $\pi(x)$, and optimizing it yields the water-filling principle. This statistical model differs from the surrogacy, latent unconfoundedness, or data fusion frameworks studied in the cited papers. (ii) We construct an orthogonal estimator that attains this efficiency bound. This estimator corrects for prediction error via a doubly-robust influence-function approach that, to the best of our knowledge, has not appeared in the prior literature on combining experimental and observational data.

Third, we extend the framework to an adaptive setting where both the predictor and the data collection design are updated sequentially as new labels arrive. We show that, under standard regularity conditions, the adaptive procedure retains the same efficiency guarantee as the static oracle design. This adaptive extension is absent from the cited experimental/observational data combination literature, and it makes the framework substantially more practical, since researchers rarely know the prediction quality across subpopulations before data collection begins.

We have revised the introduction and related work section to make these distinctions explicit. We are grateful to the reviewer for pushing us to sharpen the positioning.

Response to Reviewer 2 for MS-OPM-2025-05030

We sincerely thank Reviewer 2 for the positive assessment and the constructive suggestions. Below we respond point by point.

R2.1: Overall assessment, summary, and contribution.

Overall, I believe this is a great piece of work that only needs light revision. Congratulations to the authors for such a great paper.

[Summary] This paper addresses the challenge of costly data collection in operations management and economics by proposing a framework to integrate low-cost, potentially biased external predictions (e.g., from Large Language Models) into statistical estimation. The authors investigate how to optimally collect data to estimate population means and average treatment effects (ATE) while minimizing resource expenditure. The core contribution is the development of the INSPIRE (INtelligent SamPLing and measuREment) design, a two-stage framework that jointly optimizes the sampling density—determining which units to draw from the population—and the measurement probability—deciding whether to incur a fixed cost to observe the true label or rely solely on the free external prediction. The key theoretical argument is that optimal efficiency is achieved through a “water-filling” principle based on prediction accuracy. The authors derive the semiparametric efficiency lower bound for regular estimators in this setting, demonstrating that sampling and ground-truth measurement should be concentrated in subpopulations where the prediction residual variance is high. In other words, the design relies on cheap predictions where the model is reliable and allocates the labeling budget to areas where the model performs poorly. The paper constructs an estimator that achieves the efficiency lower bound and remains consistent even when external predictions are systematically biased. The authors extend this to an adaptive setting where the design and predictor are iteratively updated. Empirical validations on survey data suggest the method can reduce labeling costs by 40–50% without sacrificing statistical precision.

[Contribution] The research question of how to leverage cheap, potentially biased external predictions to minimize data collection costs while maintaining statistical consistency and efficiency, is clearly defined and highly relevant to the practice. Although I can see many ideas are originally from the past literature, combining all these ideas together to develop a well-executed solution to a cutting-edge problem is a significant contribution to the literature.

Our Response:

We sincerely thank the reviewer for the generous assessment, the thorough summary, and the recognition of our contribution. The reviewer’s characterization of the paper is precise, and we are encouraged by the positive evaluation. The constructive suggestions below have helped us improve the manuscript, and we address each of them in detail.

R2.2: Method, potential impact, and writing quality.

The modeling framework is quite general and flexible. The results are convincing. As AB test with the help of LLM agents has become trending in industry, I believe this paper would have significant impact. The writing is overall clear and logical. I particularly commend on their discussion after each theorem, which provide nice intuition for each result.

Our Response:

We appreciate these kind words. We are glad that the reviewer finds the framework general and the results convincing. We will continue to maintain the intuitive discussion paragraphs after each theorem, as they seem to be a valued feature of the exposition.

R2.3: Typos and proofreading.

However, there are many typos in the paper, and I ask the authors to proofread carefully. For example, on page 4 it should be “We call the INSPIRE design (IN intelligent SamPIIng and measuREment design)”. In many places the periods are missing (e.g., the first sentence of page 8).

Our Response:

We have proofread the manuscript carefully and corrected all identified typos. The two examples the reviewer flagged have been fixed: the INSPIRE acronym expansion on page 4 now reads “INtelligent SamPIIng and measuREment design,” and the missing period at the start of page 8 has been restored. Beyond these two cases, we performed a systematic pass over the full manuscript and corrected additional punctuation, spacing, and wording issues throughout.

R2.4: Improve narrative on bias vanishing.

For the narratives, when I read the paper for the first time, I had a hard time when reading Section 3: how to extend the consistent estimator concept into the case where the LLM predictor must have bias? Later on I realize that the bias will vanish as T goes to infinity. To improve the narratives, maybe the authors could emphasize in Section 2 that the predictor F can be biased, and explain after Proposition 3 why the bias vanishes.

Our Response: Thank you for this helpful suggestion. We agree that the narrative around bias management deserved clarification, and we have substantially revised Sections 2 and 3 accordingly.

In Section 2.1, we now explicitly state that the predictor f may be systematically biased, and that our estimation procedure corrects for any such bias using the subset of data for which true labels are collected. We also note that the accuracy of f affects the magnitude of the efficiency gain but not the validity of the estimator.

In Section 3, we made three changes to improve the narrative. First, we reorganized the presentation: the efficiency bound and the efficient estimator are now presented together in Section 3.1, followed by the optimal design in Section 3.2. In the previous version, the optimal design appeared before the estimator, which made it harder for the reader to see how bias is handled before encountering the design optimization. The new order allows us to establish unbiasedness and efficiency of the estimator first, and then proceed to the design problem with a clearer foundation.

Second, before Proposition 3 (the efficiency bound), we added a discussion of the missing-at-random (MAR) structure underlying our data collection scheme and introduced a simple inverse probability weighted (IPW) estimator that is unbiased regardless of the predictor quality. This makes explicit, early on, that bias correction is feasible because the measurement probability is controlled by the researcher.

Third, after establishing unbiasedness of our proposed estimator (Proposition 5), we expanded the discussion to explain why this property holds: the measurement probability $\pi(\mathbf{x})$ is known by design and plays the role of a known propensity score in the doubly-robust framework. As a result, even if the predictor is arbitrarily biased, the influence-function correction ensures unbiased estimation. As T grows and the nuisance models converge to their population counterparts, the bias correction becomes increasingly precise, and the estimator achieves the efficiency bound.

R2.5: Empirical case study for ATE / A-B testing.

Although the paper is already very good, I believe the paper can be even better if they include an empirical case study for average treatment effect with the help of LLM agents. This is because AB tests remain the most important context in practice.

Our Response:

We have added Section 5.2 in the revised manuscript, which provides an empirical illustration of the ATE extension using the Hillstrom email marketing dataset. The dataset records a three-arm RCT with 64,000 customers randomized to a men’s email campaign, a women’s email campaign, or a control (no email). Outcomes are binary visit and continuous spend. The predictor is a gradient boosted model trained on eight customer covariates.

Table 4 reports INSPIRE-ATE versus three baselines (Uniform, Stratified, Neyman) for all eight experimental conditions (two outcomes $\times$ two treatment contrasts $\times$ two budget levels). INSPIRE-ATE achieves strictly lower MSE than every baseline in every condition. The average MSE reduction is 13.3% for INSPIRE-ATE, compared with 4.2% for the Stratified/Neyman alternatives. The spend outcome is particularly instructive: stratifying on raw demographic covariates yields little or no MSE reduction (baselines range from -2.1% to $+3.8\%$), while INSPIRE’s prediction-guided design yields reductions of 9.0% to 19.9%. This illustrates the core mechanism of the paper: when covariate-based strata are coarse proxies for outcome variability, prediction residual variance provides a much sharper signal for where to concentrate the labeling budget.

R2.6: Tree-model case study may not be attractive.

On the other hand, the first case study on average income with tree model may not be attractive to readers.

Our Response:

We agree with the reviewer’s assessment. The Census income experiment with a decision-tree predictor is a serviceable proof of concept but lacks direct OM relevance. In the revised manuscript, we have moved it to Appendix A and replaced it in the main text with two more compelling examples: Section 5.1, the public-opinion survey with GPT-4o-mini (retained and expanded), and Section 5.2, the new Hillstrom email marketing ATE case study described in our response to R2.5. The Hillstrom dataset is drawn directly from retail operations (customer targeting and email campaign design), which should resonate with the OM audience.

Response to Reviewer 3 for MS-OPM-2025-05030

We thank Reviewer 3 for the thoughtful and constructive comments. Below we respond point by point.

R3.1: Summary assessment.

Research question and contribution. The research question is clearly stated and timely... The paper’s main contribution is a unified framework that jointly optimizes sampling and measurement under a labeling budget. The two-stage design is intuitive and supported by both semiparametric efficiency theory and empirical evidence. Overall, the contribution is meaningful and practically relevant for large-scale data collection settings where labels are expensive.

Method. The method is well justified for the research question. The use of semiparametric efficiency theory provides a rigorous foundation for deriving optimal sampling and measurement rules. The empirical applications further demonstrate clear and practically meaningful gains over simpler approaches.

Potential impact. The results have clear managerial and practical relevance. As firms increasingly rely on LLMs and other predictive tools that produce cheap but potentially biased outputs, an important question is when human labeling remains necessary and how to allocate labeling resources efficiently. The paper offers a principled framework for balancing automated predictions with costly ground-truth data.

Writing quality. The paper is well written and clearly organized. The proposed two-stage framework is explained intuitively, and Figure 1 is particularly helpful in illustrating the framework. The theoretical results are well explained and accessible to a broad audience. Overall, the paper is logically structured and enjoyable to read.

Our Response: We sincerely thank the reviewer for the thorough reading of our paper and the positive assessment of our research question, methodology, and writing. We are encouraged by the reviewer’s recognition that the framework is both theoretically rigorous and practically relevant. Below we address the reviewer’s specific suggestions for improvement.

R3.2: Guidance on partitioning covariates X vs. W .

The design rules depend on covariates (X), while another set of observables (W) is used only through the LLM-based prediction. It would be helpful to provide more guidance on how to partition variables between (X) and (W) in practice. Is there a principled criterion (e.g., statistical or computational) for deciding which features should inform the sampling and measurement policy directly versus only through the predictor?

Our Response:

Thank you for raising this important practical question. We have added a new discussion in Appendix C of the revised manuscript that provides guidance on partitioning covariates into design covariates $\mathbf{X}$ and predictor-only covariates $\mathbf{W}$. We address the question in three parts.

1. Partitioning as a stratification problem. In our framework, both $\mathbf{X}$ and $\mathbf{W}$ are taken as input to the prediction model. The partition only determines which covariates the sampling density $p(\mathbf{x})$ and measurement probability $\pi(\mathbf{x})$ can depend on. The problem is therefore how to choose appropriate covariates $\mathbf{X}$ to stratify and sample from. This question is related to the classical problem of choosing stratification variables in survey sampling and experimental design (????).

In the oracle setting with known conditional variances, conditioning on more covariates in $\mathbf{X}$ can only reduce the efficiency bound, because any coarse design is a feasible solution within the finer design’s optimization (see Proposition 8 in Appendix C for a formal statement). This mirrors classical results in stratified sampling—finer stratification reduces variance when stratum parameters are known (Lal, Chapter 5)—and parallels the result of Rosenbaum (1984) that matched-pair designs are asymptotically optimal among all stratified randomization schemes.

In finite samples, however, the functions $\alpha(\mathbf{x})$ and $\beta(\mathbf{x})$ that enter the optimal design must be estimated nonparametrically, and their estimation rates deteriorate with the dimension of $\mathbf{X}$. Rosenbaum (1984) demonstrate a similar phenomenon for finite-sample optimal treatment effect estimation, where the computational and statistical cost grows exponentially in the covariate dimension. Therefore, the partition of covariates into $\mathbf{X}$ and $\mathbf{W}$ involves a bias-variance tradeoff: a richer $\mathbf{X}$ lowers the oracle bound but inflates the finite-sample estimation error for the design quantities.

2. High-level criteria for the $\mathbf{X}/\mathbf{W}$ partition. Before discussing statistical aspects of the problem, we first outline three structural considerations.

First, $\mathbf{X}$ should consist of low-dimensional or categorical variables. The optimal design functions $p(\mathbf{x})$ and $\pi(\mathbf{x})$ must be estimated as functions of $\mathbf{X}$; high-dimensional $\mathbf{X}$ makes this estimation impractical. Discrete demographic variables such as age group, gender, or education level are natural candidates for $\mathbf{X}$.

Second, the population distribution $q(\mathbf{x})$ of $\mathbf{X}$ must be known, because both the optimal sampling density $p^*(\mathbf{x}) \propto q(\mathbf{x}) \alpha(\mathbf{x})$ and the importance weights $q(\mathbf{x})/p(\mathbf{x})$ in the estimator require knowledge of q . Variables whose marginal distribution is available from external sources (e.g., census data) are therefore well suited for $\mathbf{X}$. Covariates whose distribution is unknown or hard to characterize, such as free-text responses or browsing histories, should remain in $\mathbf{W}$.

Third, reserve $\mathbf{W}$ for unstructured or high-dimensional covariates where modern predictors (LLMs, deep learning models) can exploit rich information without requiring explicit density estimation or stratification rules. This recommendation aligns with established practice in experimental design: stratify on the most prognostic variables and use regression adjustment for auxiliary features (Rosenbaum, 1984).

3. Data-driven selection among candidate design covariates. After narrowing the candidates using the criteria above, a practitioner is typically left with a handful of low-dimensional covariates and must decide which subset to include in $\mathbf{X}$. We propose a pilot-based procedure using importance-weighted bootstrap (see Appendix C, Section C.3 and Algorithm 2 for details). For each candidate partition S , we run B bootstrap replicates, where each replicate places the entire INSPIRE pipeline inside the bootstrap loop: (i) draw a bootstrap resample from the pilot data, (ii) re-estimate α_S , β_S and recompute the optimal design from this resample, (iii) apply importance weights $\hat{p}_S^*(\mathbf{x})/p_0(\mathbf{x})$ to correct for the mismatch between the pilot and optimal sampling distributions, (iv) simulate measurement decisions according to $\hat{\pi}_S^*$, and (v) construct the INSPIRE estimator. Because both the design estimation and the estimator construction are recomputed in every replicate, the bootstrap variance captures the full variability of the pipeline. A partition whose design quantities are hard to estimate will show large variation in the estimated design across replicates, inflating the bootstrap variance and penalizing the partition. A similar pilot-then-design approach is used in stratification trees (Rosenbaum, 1984), where first-wave data determine the optimal stratification for the second wave, and in cross-balancing methods (Rosenbaum, 1984), where sample splitting separates feature selection from estimation to preserve inferential validity.

R3.3: Empirical illustration of the ATE extension.

The extension to average treatment effect estimation is very interesting and broadens the scope of the framework. An empirical illustration of this causal component would strengthen the paper and help readers better understand how the proposed design performs in treatment-effect applications.

Our Response:

We have added Section 5.2 in the revised manuscript, providing an empirical illustration of the ATE extension. The case study uses the Hillstrom email marketing RCT (64,000 customers, three-arm design), a publicly available dataset that is well known in the marketing and OM literature on customer targeting. Outcomes are binary visit and continuous spend; the predictor is a gradient boosted model trained on eight customer covariates.

Table 4 reports results for all eight experimental conditions. INSPIRE-ATE achieves strictly lower MSE than all baselines in every condition, with an average reduction of 13.3% versus 4.2% for Stratified/Neyman allocation. The retail marketing context makes the OM connection explicit: the firm faces a budget for sending targeted offers or collecting detailed purchase data, and INSPIRE guides allocation toward customer segments where its outcome predictions are least accurate and therefore where additional measurement is most valuable.

R3.4: Robustness checks with alternative predictive models.

The empirical evaluation (Section 5.2) relies on a single LLM (GPT-5-mini) for generating predictions. It would be useful to include robustness checks using alternative predictive models, to assess how sensitive the gains from the proposed design are to the quality and type of predictor.

Our Response:

We have added Section 5.1.3 in the revised manuscript, which reports robustness checks across five predictor types: GPT-4o-mini (the original LLM), DeepSeek-V3.2, XGBoost trained on the survey covariates, a Population Mean predictor (the grand mean of labels collected so far), and a Random predictor (white noise). The results are reported in Table 3 for both datasets and two budget levels.

Two findings stand out. First, INSPIRE reduces MSE by 37–47% across all five predictors and both datasets. Even the Random predictor, which carries no information about individual outcomes, yields a 37% MSE reduction; this reduction comes entirely from the optimal sampling component of the design, which concentrates observations where the population distribution is most informative. Second, the gap between the best LLM (GPT-4o-mini, DeepSeek-V3.2) and the Random predictor is approximately 9 percentage points, reflecting the additional efficiency gain from an accurate conditional mean estimate. Together, these results show that the framework is robust to predictor quality: a researcher who lacks access to a strong LLM still benefits substantially from INSPIRE, and a researcher with a high-quality predictor gains more.

We note that Reviewer 3 refers to “GPT-5-mini” in the original comment; we believe this is a typographical error for GPT-4o-mini, which is the model used in our experiments.

R3.5: Practical guidance for early-stage data collection.

The performance of the method appears to depend on accurate estimation of the components that enter the optimal design (e.g., those in Theorem 1). It would be helpful to discuss practical guidance for the early stages of data collection, when these estimates may be very noisy.

Our Response:

Thank you for this important question. We have added a discussion of early-stage practical guidance in Section 3.3 of the revised manuscript. The short answer is that noisy estimates of the design quantities affect only the *efficiency* of the estimator, never its *validity*, and simple safeguards can limit the efficiency loss in early rounds.

The key theoretical reassurance is that the INSPIRE estimator is unbiased regardless of the quality of the estimated design. As stated in Propositions 4 and 5, $\mathbb{E}[\hat{\theta}_T] = \theta$ holds for *any* choice of design (p, π) and *any* nuisance models $\hat{\mu}$ and $\hat{\tau}$, without requiring that these be close to their population counterparts. A suboptimal design increases the variance of the estimator but does not introduce bias. This property is inherited from the doubly-robust structure of the influence function and is analogous to the well-known result in semiparametric estimation that augmented IPW estimators remain consistent even with misspecified nuisance models (?).

Given this guarantee, our practical recommendation for early-stage data collection is to use a *mixture design* that blends the estimated optimal design with a uniform baseline. Specifically, we suggest using

$$\tilde{p}(\mathbf{x}) = (1 - \epsilon) \hat{p}^*(\mathbf{x}) + \epsilon q(\mathbf{x}), \quad \tilde{\pi}(\mathbf{x}) = (1 - \epsilon) \hat{\pi}^*(\mathbf{x}) + \epsilon \gamma,$$

where $\epsilon \in (0, 1]$ is a mixing parameter that starts large (e.g., $\epsilon = 0.5$) and decreases as estimates improve. This mixture provides two benefits. First, it guarantees a lower bound on the sampling and measurement probabilities for every subgroup, preventing the extreme importance weights $q(\mathbf{x})/p(\mathbf{x})$ that can arise when the estimated optimal design assigns very low probability to some region. Second, the uniform component ensures broad coverage in early rounds, collecting data from all subgroups and thereby improving the estimates of $\alpha(\mathbf{x})$ and $\beta(\mathbf{x})$ for subsequent rounds.

Two additional recommendations complement the mixture design. First, if domain knowledge about prediction quality across subgroups is available—for instance, from a related study or from the predictor’s known strengths and weaknesses—it can be used to warm-start the estimates of $\hat{\alpha}$ and $\hat{\beta}$, reducing the number of rounds needed before the design converges. Second, in connection with the covariate partition guidance discussed in our response to R3.2, we recommend starting with a coarser partition (fewer design covariates $\mathbf{X}$) in early stages, since coarser designs require fewer observations to estimate reliably.

Finally, we note that the simulation results in Appendix A (Figure 2) provide empirical evidence that the estimation cost in early stages is small. The adaptive INSPIRE algorithm, which starts with no knowledge of α and β and estimates them on the fly, achieves MSE very close to the INSPIRE-Oracle benchmark that uses the true distributional information throughout. This suggests that the adaptive procedure converges quickly to a near-optimal design, and the efficiency loss from noisy early-stage estimates is modest in practice.

R3.6: Clarify technical challenges relative to standard semiparametric efficiency analysis.

The theoretical sections are rigorous. However, the paper would benefit from a clearer high-level discussion of the main technical challenges and methodological innovations, especially in relation to standard semiparametric efficiency analysis.

Our Response:

Thank you for this helpful suggestion. We agree that the previous version emphasized the formal derivations more than the high-level technical message. In the revised manuscript, we now clarify more explicitly which parts of our analysis follow standard semiparametric arguments and which parts are specific to our setting.

At a high level, the fixed-design i.i.d. derivation builds on standard semiparametric efficiency tools for missing-at-random models. The main methodological novelty is not a new generic efficient influence function calculus, but rather the way these tools must be adapted to our design problem. First, unlike standard semiparametric settings where the observed-data law is taken as fixed, here both the sampling density $p(x)$ and the measurement probability $\pi(x)$ are design variables chosen by the researcher, while the

estimand remains defined under the target population distribution $q(x)$. As a result, the efficient influence function and the efficiency bound must jointly account for both covariate shift through $q(x)/p(x)$ and design-induced missingness through $\pi(x)$. This is what makes the lower bound a function of the design itself and allows it to serve as the basis for optimizing data collection.

Second, the external prediction F is observed for every sampled unit but may be systematically biased. The technical challenge is therefore to use F for efficiency gain without requiring it to be correct. Our solution is to treat F as an always-observed auxiliary variable and derive an orthogonal estimator based on the nuisance functions $\mu(x) = \mathbb{E}[Y \mid X = x]$ and $\tau(x, f) = \mathbb{E}[Y \mid X = x, F = f]$. This yields an estimator that remains unbiased even under arbitrary predictor bias and attains the semiparametric efficiency bound.

Third, the adaptive extension is nonstandard from a semiparametric perspective because the design and the predictor are updated from past data, so the observations are no longer i.i.d. To handle this, we exploit the filtration structure of the sequential procedure: at time t , all estimated models and design rules are $\mathcal{F}_{t-1}$ -measurable, which creates a martingale-difference structure analogous to cross-fitting. This allows us to recover the same asymptotic variance as in the static oracle design and to derive a regret bound that quantifies the cost of estimating the design online.

We have added a short high-level discussion to the theoretical section to make these points explicit and to better position our contribution relative to standard semiparametric efficiency analysis.